Vantara Customer Intelligence

Customer behaviour prediction platform: final report. Data: Online Retail II (UCI), 1 Dec 2009 to 9 Dec 2011. Scores as of 2011-09-11 (training snapshot) and 10 Dec 2011 (live).

Churn ROC-AUC	Churn recall	CLV R^2 (test)	Segments
0.811	**87%**	**0.69**	**6**

1. Summary

The platform scores every customer seen in the last two years for 90-day churn risk, 12-month lifetime value, next-purchase probability, unusual spending and segment. Each churn score has a SHAP explanation, LIME is computed on demand, and a plain-language sentence explains every flag. Results are stored in PostgreSQL and served by a FastAPI service to a React dashboard. The full stack starts with `docker-compose up`, and the pipeline re-runs end-to-end with `python -m src.pipeline`.

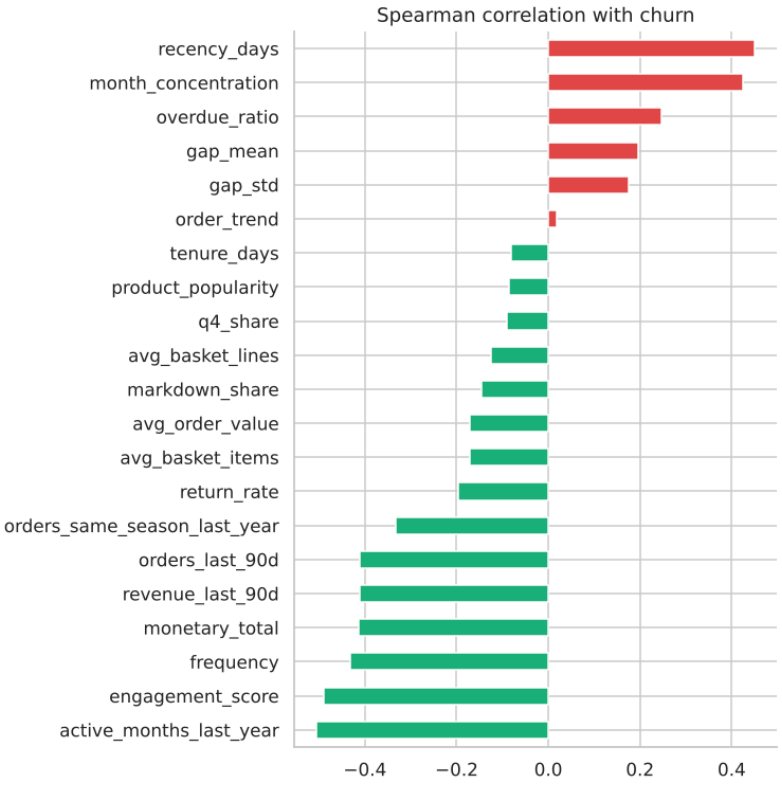
2. Success metrics (PRD §3.2)

Metric	Target	Result	Status
Churn ROC-AUC (test)	≥ 0.80	0.811	✅ Met
CLV R^2 (test)	≥ 0.60	0.692	✅ Met
Churn recall	≥ 0.70	0.874	✅ Met
API p95, single prediction	< 400 ms	≈ 140 ms (asserted < 400 ms in tests)	✅ Met
Dashboard full segment view	< 3 s	≈ 1 s on a local stack	✅ Met
Single-command pipeline	100% scripted	`python -m src.pipeline`	✅ Met

3. Data and features

Both workbook sheets are merged and cleaned. The cleaning rules: exact duplicates are removed after descriptions are standardised per StockCode; non-positive prices are dropped as adjustments; extreme orders that were fully cancelled are removed as entry errors; administrative codes (postage, fees, manual entries) are flagged and excluded from product features; and returns are kept as a signal. A pandera schema blocks feature engineering if any gate fails. Of the 1,027,003 clean lines, 22% have no Customer ID and are used only at product level.

Features are computed at a cutoff date using only earlier transactions; tests prove that deleting or altering later data leaves every feature unchanged. Labels: churn means no purchase in the 90 days after the cutoff (churn rate 56%); CLV means net revenue in the 365 days after a separate cutoff.



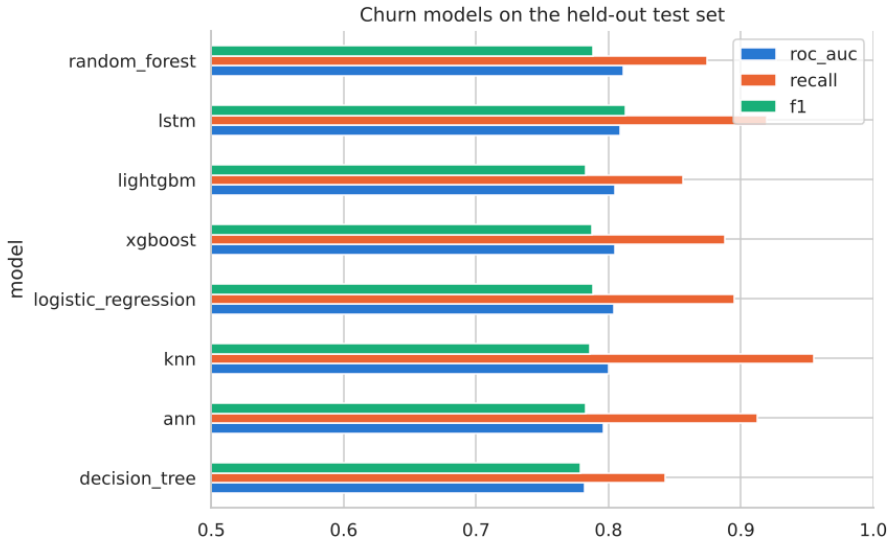
4. Churn models

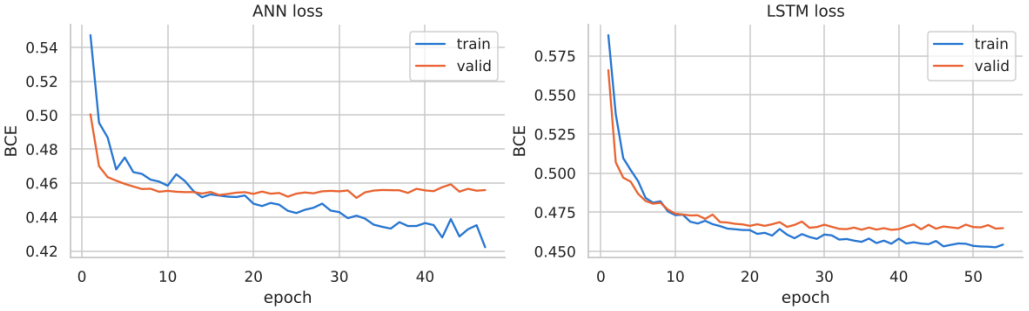
Split 70/15/15, stratified, seed 42 (3679/788/789 customers). Hyperparameters were tuned by randomised search with stratified 5-fold CV on the training set; XGBoost then used early stopping on validation. Each model's alert threshold is the highest-F1 cutoff with validation recall ≥ 70%. The test set was used once. All 12 runs are logged in `models_artifacts/runs.jsonl`.

Model	ROC-AUC	Recall	Precision	F1	Accuracy	CV AUC
Random forest	**0.811**	**0.874**	**0.718**	**0.789**	**0.735**	**0.810**
LSTM (purchase sequences)	0.809	0.919	0.728	0.813	0.760	—
LightGBM	0.805	0.857	0.721	0.783	0.731	0.803
XGBoost	0.805	0.888	0.707	0.787	0.729	0.807
Logistic regression	0.804	0.895	0.705	0.789	0.729	0.803
k-nearest neighbours	0.800	0.955	0.668	0.786	0.706	0.804
Feed-forward ANN	0.796	0.913	0.685	0.783	0.714	—
Decision tree	0.782	0.843	0.723	0.778	0.729	0.782

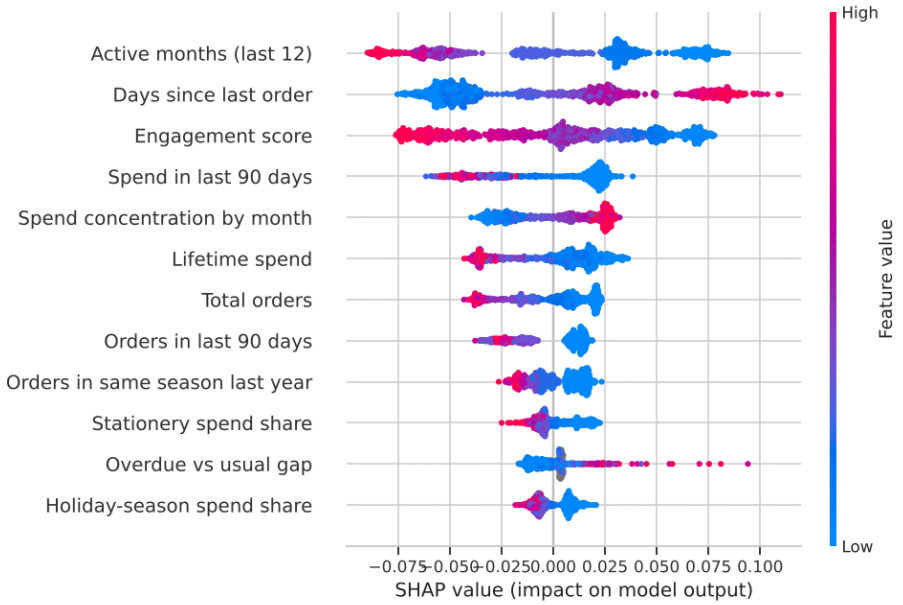
Model selection

Random forest serves production. It had the best validation ROC-AUC among the tree ensembles, and tree ensembles get exact, fast TreeSHAP explanations for every customer. The deep models are competitive but not better on ~5.8k customers, as PRD §19 anticipated. Its test errors correlate with the forest's at 0.79 (ANN: 0.77), so the deep models mostly make the same mistakes and add little as an independent check. The LSTM is kept because it models order timing directly, and it supplies the dashboard's next-purchase probability. On customers active in the last year, a harder subset, the production model scores 0.805 ROC-AUC.

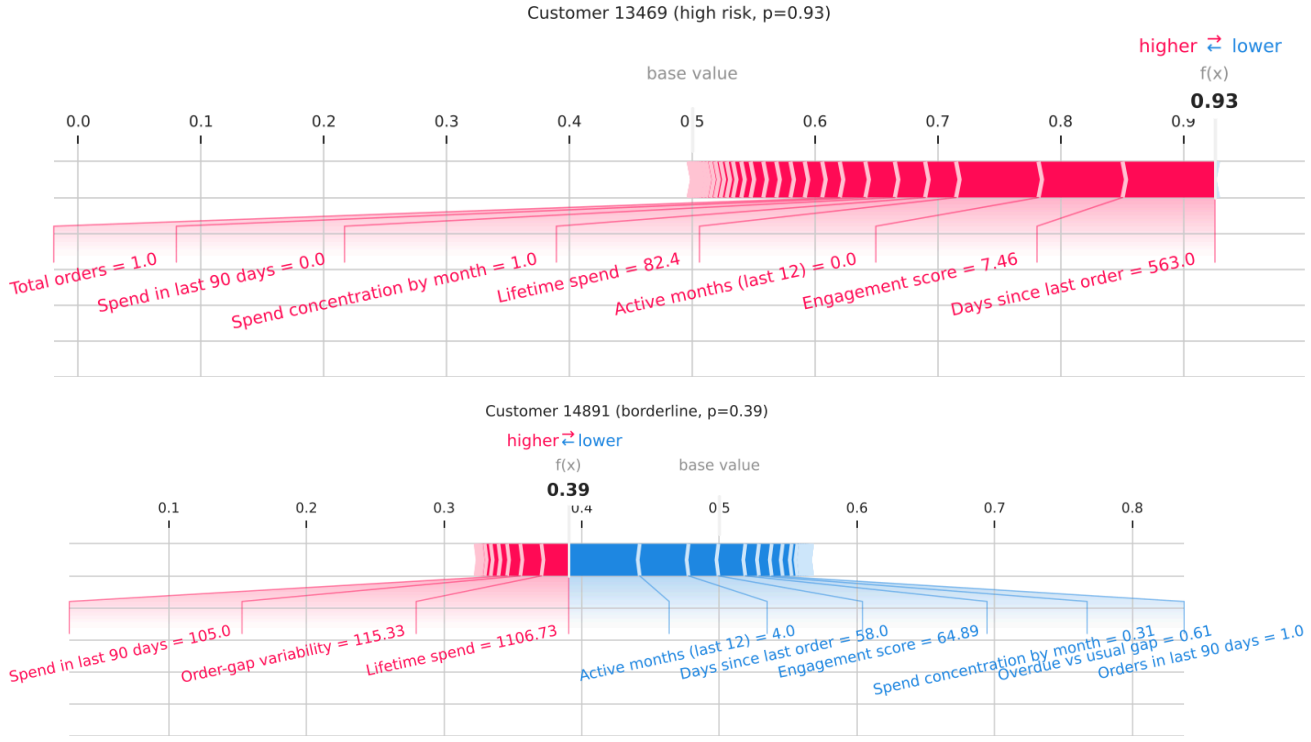

Churn models on the held-out test set
random_forest
lstm
lightgbm
xgboost
logistic_regression
knn
ann
decision_tree
model
roc_auc
recall
f1
0.5
0.6
0.7
0.8
0.9
1.0

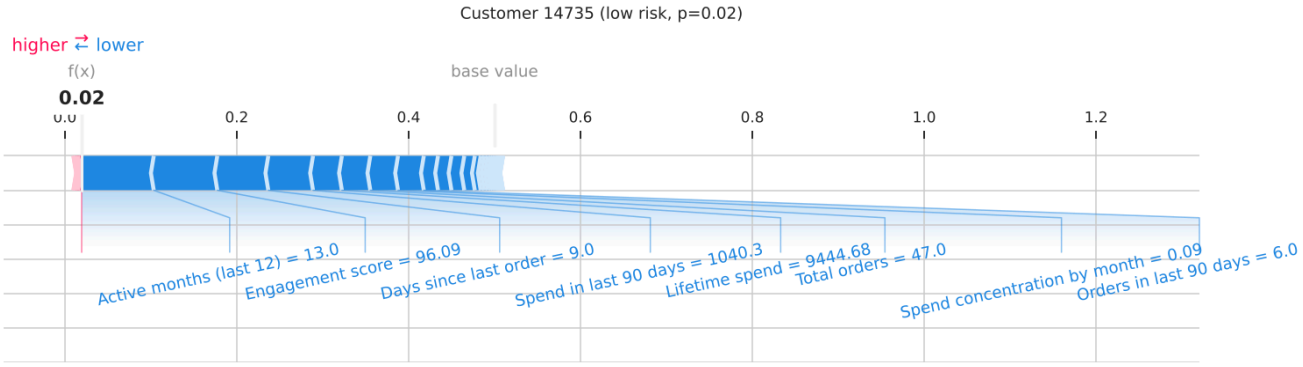

ANN loss
train
valid
BCE
epoch
LSTM loss
train
valid
BCE
epoch

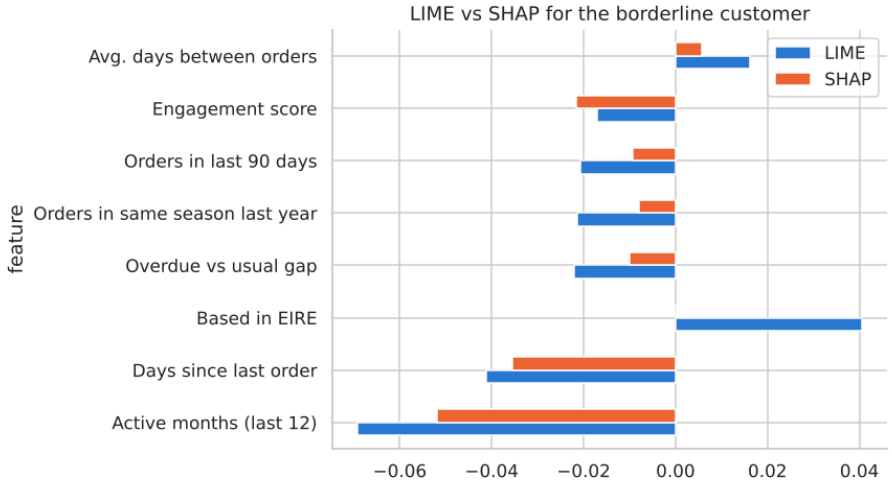
5. Explainability, segments, CLV and anomalies

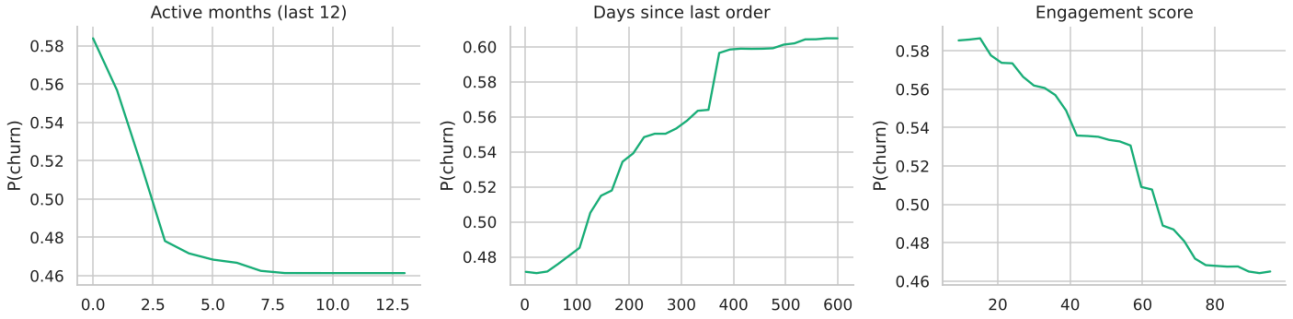


Force plots for a low-risk, a high-risk and a borderline customer:



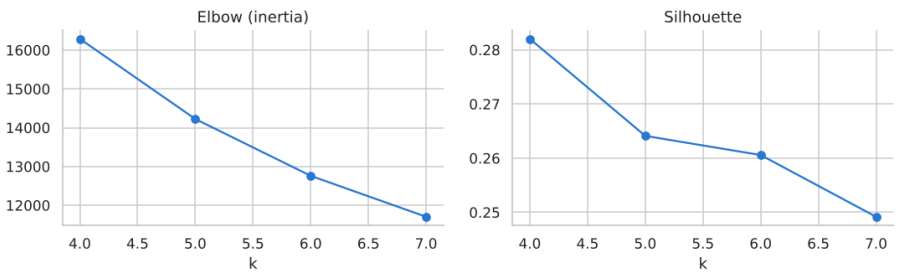






Segmentation

K-Means ran on log-scaled recency, frequency, spend, order value, Q4 share and tenure. The rule for k: take the most granular k whose silhouette is within 0.025 of the best and whose smallest segment holds ≥ 3% of customers. That gave k = 6 (silhouette 0.26, Davies-Bouldin 1.21). A Gaussian mixture (7 components by BIC) scored silhouette 0.13 and ARI 0.34 against K-Means. The mixture's overlapping components do not separate customers better, so K-Means is kept. Clusters get business personas through an optimal assignment on their standardised centroids.



Customer lifetime value

Candidate	CV R^2	Validation R^2	Validation MAE
ridge	0.522	0.880	£1,105
lightgbm_tweedie	0.559	0.610	£1,042
lightgbm_log	0.297	0.313	£1,136

The selected model (lightgbm_tweedie) scores test R^2 0.69, MAE £889 and RMSE £3,311. Future value is extremely skewed: a few wholesalers dominate the variance. A Tweedie loss handles the many zero-revenue customers and the long tail together. R^2 varies noticeably between folds, so the CV figure is the more conservative estimate.

Anomaly detection

A 3-unit bottleneck autoencoder reconstructs log-scaled spending features. Accounts whose reconstruction error is above the validation 99th percentile are flagged: 1.4% of test customers. Their median lifetime spend is £1,373 against £743 overall, so they lean toward larger, irregular accounts that are worth a manual look.

6. Limitations and discussion

- Scope: all customers seen in the last two years are scored, matching the PRD's ~5.9k base. Long-lapsed customers are easy to call; on recently active customers the ROC-AUC is 0.805.
- The churn window (Sep–Dec) covers the Christmas peak, so seasonal features matter more than they would in spring. Models should be retrained each quarter.
- CLV accuracy depends on a handful of very large accounts. Use the value tier (High / Mid / Low) rather than the exact £ figure for small customers.
- Class imbalance is mild here (≈56% churn), so class weights were used instead of SMOTE.

7. Future enhancements

- Transformer sequence model once more sequence history is available.
- Streaming ingestion and daily rescoring.
- Authentication, role-based access and CI/CD.
- A/B tests to measure the uplift of retention campaigns driven by these scores.
- Uplift modelling to target customers who respond to offers, not just those at risk.

Appendix A. Mathematical foundations

A.1 Statistics and probability

The engagement score averages percentile ranks: $E = 100 \cdot \tfrac{1}{3}\,[\hat{F}_{-R}(r) + \hat{F}_F(f) + \hat{F}_M(m)]$, where $\hat{F}$ is the empirical CDF of the reference population. Classifiers estimate the conditional probability $P(\text{purchase} \mid R, F, M, \ldots)$ directly through their predicted probabilities.

A.2 Linear algebra and PCA

The segment feature matrix $X \in \mathbb{R}^{n \times d}$ is log-transformed and standardised. PCA eigendecomposes the covariance matrix:

> $$\Sigma = (1/n)\, X^{\mathsf{T}}X, \quad \Sigma\, v_k = \lambda_k\, v_k, \quad z = [v_1\ v_2]^{\mathsf{T}}\, x$$

The two leading eigenvectors give the dashboard's customer map.

A.3 Logistic regression and cross-entropy

> $$P(y{=}1 \mid x) = \sigma(w^{\mathsf{T}}x + b) = 1 \,/\, (1 + e^{-(w^{\mathsf{T}}x + b)})$$

> $$L(w) = -(1/n)\, \Sigma_i\, [y_i \log \hat{y}_i + (1{-}y_i) \log(1{-}\hat{y}_i)] + \lambda\, [\rho \lVert w \rVert_1 + (1{-}\rho)/2\, \lVert w \rVert_2{}^2]$$

The elastic-net mix $\rho \in \{0, 0.5, 1\}$ and the inverse strength $C = 1/\lambda$ are tuned by cross-validation (saga solver). Class weights $n/(2n_c)$ rebalance the loss.

A.4 Gradient descent and backpropagation

> $$w \leftarrow w - \eta\, \nabla L(w) \quad (\text{Adam: } m \leftarrow \beta_1 m + (1{-}\beta_1) g,\ v \leftarrow \beta_2 v + (1{-}\beta_2) g^2,\ w \leftarrow w - \eta\, \hat{m} \,/\, (\sqrt{\hat{v}} + \varepsilon))$$

Backpropagation applies the chain rule layer by layer: $\partial L/\partial W^{(l)} = \delta^{(l)}\, a^{(l-1)\mathsf{T}}$, with $\delta^{(l)} = (W^{(l+1)\mathsf{T}}\, \delta^{(l+1)}) \odot f'(z^{(l)})$. In the LSTM, gradients flow through the cell state $c_t = f_t \odot c_{t-1} + i_t \odot g_t$. The loss curves in §4 confirm convergence; early stopping (patience 15) restores the epoch with the best validation loss.

A.5 Regularisation

L1/L2 penalties are applied in the logistic regression (above). The networks use weight decay 10^{-4} (an L2 penalty), batch normalisation, and dropout $p = 0.3$ (ANN) or $p = 0.2$ (LSTM), which zeroes units with probability p during training.

A.6 Evaluation metrics

> $$\text{Precision} = \text{TP}/(\text{TP}{+}\text{FP}),\ \text{Recall} = \text{TP}/(\text{TP}{+}\text{FN}),\ \text{F1} = 2\text{PR}/(\text{P}{+}\text{R})$$

$$\text{ROC-AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(u))\,du = P(\hat{s}(x^+) > \hat{s}(x^-))$$

$$R^2 = 1 - \Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2, \quad \text{MAE} = (1/n)\Sigma|y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{((1/n)\Sigma(y_i - \hat{y}_i)^2)}$$

$$\text{Silhouette } s(i) = (b(i) - a(i)) / \max(a(i), b(i)); \quad \text{Davies-Bouldin} = (1/k)\,\Sigma_i \max_{j \neq i} (\sigma_i + \sigma_j)/d(c_i, c_j)$$

A.7 SHAP and LIME

$$\varphi_j = \Sigma_{S \subseteq F \setminus \{j\}} |S|!(|F| - |S| - 1)!/|F|! \cdot [f(S \cup \{j\}) - f(S)], \quad f(x) = \varphi_0 + \Sigma_j \varphi_j$$

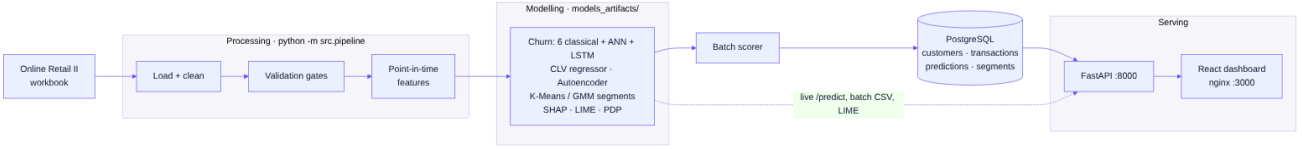
LIME fits a weighted sparse linear model g in a neighbourhood π_x: $\text{argmin}_g\, L(f, g, \pi_x) + \Omega(g)$.

A.8 Tweedie loss (CLV)

$$-\log L \propto \Sigma_i \left[-y_i \mu_i^{1-p}/(1-p) + \mu_i^{2-p}/(2-p) \right], \quad 1 < p < 2$$

This is a compound Poisson-gamma model: it puts a probability mass at exactly £0 and has a continuous right-skewed tail.

Appendix B. Architecture and data model



customers			
bigint	customer_id	PK	
text	country		
float	recency_days		
float	frequency		
float	monetary_total		
float	engagement_score		
date	first_purchase		
date	last_purchase		
float	cat_shares		one column per category

segments		
text	segment	PK
int	size	
float	churn_rate	
float	avg_clv	
float	revenue_at_risk	
text	action	

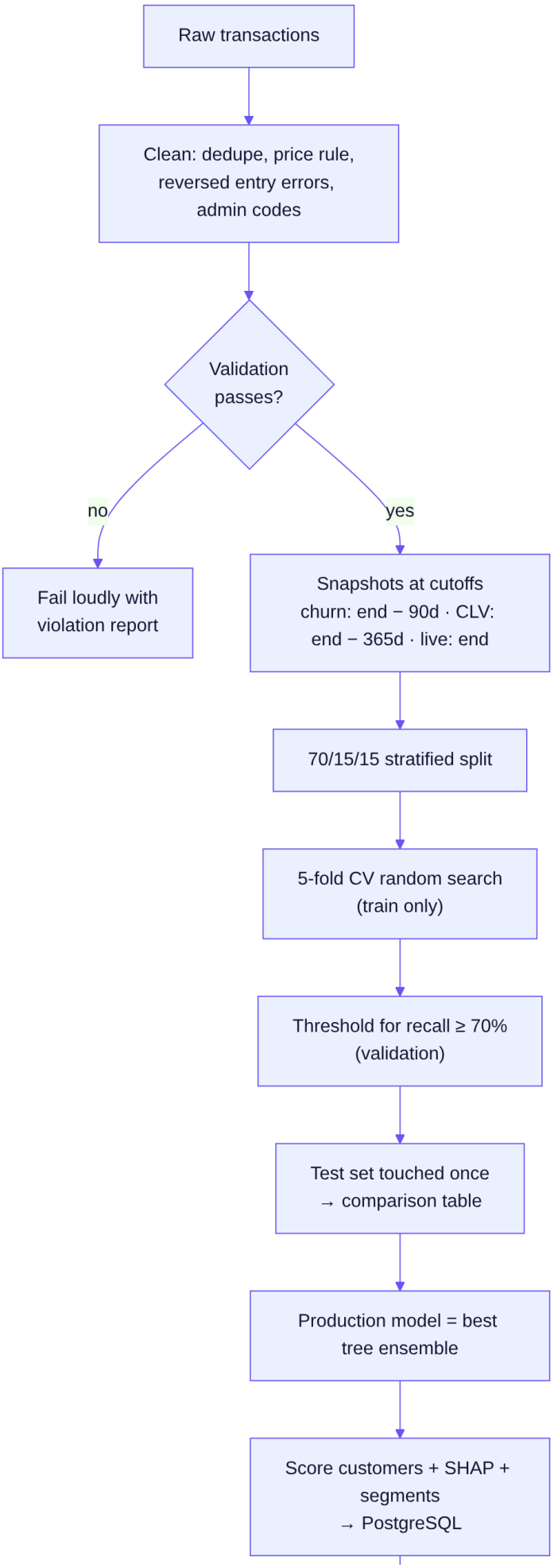
places

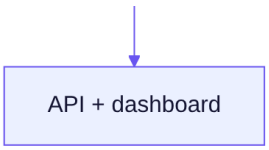
groups

scored as

transactions		
bigint	customer_id	FK
text	invoice	
timestamp	invoice_date	
text	month	
text	stock_code	
text	description	
text	category	
int	quantity	
float	price	
float	revenue	
bool	is_return	
bool	is_admin	

predictions		
bigint	customer_id	PK
float	churn_prob	
text	risk_band	
float	clv_pred	
float	revenue_at_risk	
float	next_purchase_prob	
bool	is_anomaly	
text	segment	FK
float	pca_x	
float	pca_y	
json	shap	
text	explanation	
json	recommendations	
float	prev_churn_prob	
date	scored_at	


Raw transactions
Clean: dedupe, price rule, reversed entry errors, admin codes
Validation passes?
no
yes
Fail loudly with violation report
Snapshots at cutoffs churn: end − 90d · CLV: end − 365d · live: end
70/15/15 stratified split
5-fold CV random search (train only)
Threshold for recall ≥ 70% (validation)
Test set touched once → comparison table
Production model = best tree ensemble
Score customers + SHAP + segments → PostgreSQL


API + dashboard